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Portable electronic device

Abstract

A portable electronic device includes a housing, a heat-dissipation component, a bracket, a door structure, a driving mechanism, and a driven linkage. The housing includes a heat-dissipation opening disposed in the housing. The bracket is disposed in the housing and surrounds the heat-dissipation component. The door structure is configured to move between a closed position covering the heat-dissipation opening and an open position exposing the heat-dissipation opening. The driving mechanism is coupled between the bracket and the door structure to drive the door structure to rotate and move. The driven linkage is coupled between the bracket and the door structure. When the door structure is driven to rotate and move, the door structure drives the driven linkage to rotate and move, so that the driven linkage and the driving mechanism jointly drive the door structure to move between the closed position and the open position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims the priority benefit of Taiwan application serial no. 111140856, filed on Oct. 27, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.
BACKGROUND
Technical Field
(2) The present disclosure generally relates to a portable electronic device.
Description of Related Art
(3) In recent years, the high development of mobile communication technology has created many novel applications. At the same time, various new types of portable electronic devices have been developed, and through continuous improvement, the central processing unit of the current portable electronic devices has considerable computing ability, and can support operation of a variety of software with excellent convenience of use. However, generally speaking, for the portability and convenience of use, on portable electronic devices such as smart phones, tablet computers or wearable devices, the fans are relatively large in size and generate a certain amount of noise during operation. In the past, it was rare to see a design integrating a fan into a portable electronic device. Therefore, in order to maintain CPU of the portable electronic device to operate with higher power and performance, it is necessary to provide additional cooling for the portable device.
SUMMARY
(4) Accordingly, the present disclosure provides a portable electronic device including a housing, a heat dissipation component, a bracket, a door structure, a driving mechanism, and a driven linkage. The housing includes a heat dissipation opening. The heat dissipation component is disposed in the housing and corresponds to the heat dissipation opening. The bracket is disposed in the housing and surrounding the heat dissipation component. The door structure is configured to rotate and move between a closed position covering the heat dissipation opening and an open position exposing the heat dissipation opening. The driving mechanism is pivotally connected between the bracket and the door structure to drive the door structure to rotate and move. The driven linkage is coupled between the bracket and the door structure, so as to drive the driven linkage to rotate and move when the door structure is driven to rotate and move, such that the driving mechanism and the driven linkage jointly drive the door structure to rotate and move between the closed position and the open position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.
(2) FIG. **1** is a schematic view of a portable electronic device according to an embodiment of the disclosure.
(3) FIG. **2** is a schematic view of a portable electronic device integrating with a fan module according to an embodiment of the present disclosure.

(4) FIG. **3** is a schematic cross-sectional view of the portable electronic device integrating with the fan module in FIG. **2**.
(5) FIG. **4** is a schematic view of a door structure of a portable electronic device at a closed position according to an embodiment of the disclosure.
(6) FIG. **5** is a schematic side view of the door structure of the portable electronic device in FIG. **4** at the closed position.
(7) FIG. **6** is a schematic view of the door structure of the portable electronic device in FIG. **4** at an open position.
(8) FIG. **7** is a schematic view of a door structure of a portable electronic device at a closed position according to another embodiment of the present disclosure.
(9) FIG. **8** is a schematic cross-sectional view of the door structure of the portable electronic device in FIG. **7** at a closed position.
(10) FIG. **9** is a schematic view of the door structure of the portable electronic device in FIG. **7** at an open position.
(11) FIG. **10** is a schematic cross-sectional view of the door structure of the portable electronic device in FIG. **9** at the open position.

DESCRIPTION OF THE EMBODIMENTS

(12) The present invention will now be described more fully with reference to the accompanying drawings, in which exemplary embodiments of the invention are shown. The terms used herein such as “above”, “below”, “front”, “back”, “left” and “right” are for the purpose of describing directions in the figures only and are not intended to be limiting of the invention. Moreover, in the following embodiments, the same or similar devices are denoted by the same or similar referential numbers.

(13) The content of this disclosure will be described in detail below with different figures. Please refer to FIG. **1** to FIG. **3**, in some embodiments, a portable electronic device **100** may be a portable electronic device such as a smart phone, a tablet computer, or a wearable device. The portable electronic device **100** includes a housing **110**, a heating component **120**, a heat dissipation component **130**, a bracket **140**, a door structure **150**, a driving mechanism **160** and a driven linkage **170**. The housing **110** can be configured to define an accommodating space, and includes a heat dissipation opening **112** on its surface. In this embodiment, the heat dissipation opening **112** is located at a back surface of the portable electronic device **100**. The heating component **120** is disposed in the accommodating space defined by the housing **110**. In some embodiments, the heating component **120** is, for example, a central processing unit (Central Processing Unit, CPU) of the portable electronic device **100** or other types of heating components, and can be configured on the circuit board in the housing **110**. The heat dissipation component **130** may include a set of heat dissipation fins, which may be thermally coupled to the heating component **120**.

(14) In other embodiments, the heat dissipation component **130** can also be disposed around the heating component **120** and the heat dissipation component **130** is connected to the heating component **120** by a component such as a vapor chamber, so that the heat dissipation component **130** can perform heat dissipation to the heating component **120**. In some embodiments, a contact surface of the heat dissipation component **130** for contacting the heating component **120** or the vapor chamber may be coated with thermal paste to enhance the heat conducting efficiency, but the present disclosure is not limited thereto. In one embodiment, the heat dissipation component (fin) **130** may be made of the same material as the housing **110**, or even the heat dissipation component **130** may be an extension part of the housing **110**. Certainly, in other embodiments, the heat dissipation component **130** may also be a separate heat dissipation (fin) element, which is fixed in the portable electronic device **100** by, for example, welding.

(15) In some embodiments, the heat dissipation component **130** is disposed in the housing **110** and corresponds to the heat dissipation opening **112**. In other words, the heat dissipation opening **112** of the housing **110** can expose at least a part of the heat dissipation component **130** (e.g., heat

dissipation fins), so as to improve the heat dissipation efficiency of the heat dissipation component **130**. In this embodiment, the heat dissipation component **130** is a passive heat dissipation component, but the disclosure is not limited thereto. The door structure **150** is configured to rotate and move between a closed position for covering the heat dissipation opening **112** and an open position for exposing the heat dissipation opening **112**.

(16) In some embodiments, the portable electronic device **100** shown in FIG. **1** can be integrated with other heat dissipation kits to further facilitate the heat dissipation of the heating component **120** and the heat dissipation component **130**. For example, in this embodiment, the portable electronic device **100** may include a fan module **200** sleeved on the housing **110** as shown in FIG. **2** and FIG. **3**. Accordingly, when heat dissipation is in greater need, the user may detachably arrange the fan module **200** on the housing **110** to integrate with the portable electronic device **100** and facilitate the heat dissipation of the portable electronic device **100**. In one embodiment, the fan module **200** may include a frame **210** and a fan **220** disposed in the frame **210**. The frame **210** can be fitted with at least a part of the housing **110** so as to be detachably disposed on the housing **110** of the portable electronic device **100**. Moreover, the frame **210** may have an opening **212** corresponding to an air inlet **222** of the fan **220** and an opening **214** corresponding to the air outlet **224** of the fan **220**. Moreover, the opening **214** of the frame **210** corresponds to the heat dissipation opening **112** of the housing **110**.

(17) In some embodiments, when the fan module **200** is sleeved on the portable electronic device **100**, the door structure **150** is at the open position as shown in FIG. **3**. At the time, the air outlet **224** of the fan **220** faces the heat dissipation component **130**, and the door structure **150** extends between the air outlet **224** and the heat dissipation component **130**. Accordingly, the door structure **150** at the open position can not only expose the heat dissipation component **130** in the housing **110**, but also provide the effect of guiding the air flow. In this embodiment, the air outlet **224** of the fan **210** faces the heat dissipation opening **112** of the housing **110**, and an acute angle θ**1** is included between a lower surface S**1** of the fan **220** and a back surface S**2** of the housing **110**. In other words, with the design and lifting of the frame **210**, the fan **210** can be inclined relatively to the back surface S**2** of the housing **110**, and the angle of inclined slope may be about the same as or similar to the angle of inclined slope of the door structure **150** at the open position, so that the cooling air of the fan **210** can be blown out from the air outlet **224** and directed toward the heat dissipation component **130** along the door structure **150**. Under such a configuration, the cooling airflow provided by the fan **220** can smoothly flow to the heat dissipation component **130** through the guidance of the door structure **150**, so as to facilitate the heat dissipation of the heat dissipation component **130**.

(18) Referring to FIG. **2** to FIG. **6**, in some embodiments, the bracket **140** is disposed in the housing **110** and surrounds the heat dissipation component **130**. For example, the bracket **140** may be locked to, for example, a bottom plate in the housing **110**, but the disclosure is not limited thereto. In this embodiment, the bracket **140** is configured to encircle the heat dissipation component **130** and define a bracket opening **142**. The bracket opening **142** corresponds to the heat dissipation opening **112** of the housing **110** to expose the heat dissipation component **130** together. In this embodiment, the door structure **150** is configured to be driven to rotate and move relative to the bracket **140**, so as to rotate and move between a closed position for covering the heat dissipation opening **112** (as the closed position shown in FIG. **4**) and an open position for exposing heat dissipation opening **112** and the heat dissipation component **130** underneath (as the open position shown in FIG. **6**).

(19) In this embodiment, the driving mechanism **160** is pivotally connected between the bracket **140** and the door structure **150** to drive the door structure **150** to rotate and move. The driven linkage **170** is coupled between the bracket **140** and the door structure **150**. Accordingly, when the door structure **150** is driven to rotate and move, the driven linkage **170** is driven to rotate and move, so that the driven linkage **170** and the driving mechanism **160** together drive the door

structure **150** to rotate and move between the closed position shown in FIG. **4** and the open position shown in FIG. **6**.

(20) Specifically, the driving mechanism **160** includes a gear set **162** and a driving linkage **164** coupled between the gear set **162** and the door structure **150**, wherein the gear set **162** is configured to be driven to rotate, so as to drive the driving linkage **164** to rotate and move, and thereby drive the door structure **150** to rotate and move. In detail, the driving mechanism **160** may further include a driving motor **166**. The gear set **162** may further include a driver gear **1621** coupled to the driving motor **166** and idle gears **1622**, **1623** coupled to the driver gear **1621**. The driving motor **166** can, for example, be coupled to the axis of the driver gear **1621** to drive the driver gear **1621** to rotate, and the idle gear **1622** can, for example, be engaged with the driver gear **1621**, and the idle gear **1623** can then, for example, be engaged with the idle gear **1622** to drive the driver gear **162** and the idle gears **1622**, **1623** to rotate in sequence. It should be noted that the gear set **162** in this embodiment includes a driver gear **1621** and two idle gears **1622**, **1623**, but the present disclosure is not limited thereto. In other embodiments, the gear set **160** can be customized according to actual needs. Configurations have more or fewer idle gears.

(21) In some embodiments, two opposite ends of the driving linkage **164** can be respectively coupled to the idle gear **1623** and the door structure **150**, so that the driving linkage **164** is driven by the idle gear **1623** to rotate, and then drive the door structure **150** to rotate and be lifted. Furthermore, the driving linkage **164** can be coupled to a first coupling point of the door structure **150**, and the driven linkage **170** can be coupled to a second coupling point of the door structure **150**, so as to jointly drive the door structure **150** to rotate and move through the two coupling points of the door structure **150**. In this way, when the door structure **150** is driven to rotate and be lifted, the door structure **150** will drive the driven linkage **170** coupled thereto to rotate along with the door structure **150**, so that the driving linkage **164** and the driven linkage **170** can together drive the door structure **150** to rotate and move, so that the door structure **150** rotates and moves between the closed position as shown in FIG. **4** and the open position as shown in FIG. **6**.

(22) Please refer to FIG. **5**, for example, in this embodiment, when the portable electronic device tends to open the door structure **150**, the driving motor **166** may be driven by a controller, so that the driver gear **1621** is driven by the driving motor **166** to rotate in the first direction R**1** such as a clockwise direction, and the idle gear **1622** is driven to rotate in an opposite second direction R**2** such as a counter-clockwise direction, thereby drive the idle gear **1623** to rotate in the first direction R**1**, e.g., clockwise direction. One end of the driving linkage **164** can be coupled to an axis of the idle gear **1623** through a screw or the like, so as to be driven by the idle gear **1623** to rotate along the first direction R**1**, e.g., clockwise direction. The first coupling point of the door structure **150** can be coupled to the other end of the driving linkage **164**, so as to rotate and be lifted along with the other end of the driving linkage **164**, and simultaneously drive the driven linkage **170** coupled thereto to rotate and be lifted together. In this way, the driving linkage **164** and the driven linkage **170** can jointly control the route of rotation and movement of the door structure **150**, so that the door structure **150** can rotate and move from the closed position as shown in FIG. **4** to the closed position as shown in FIG. **6**.

(23) Similarly, when the portable electronic device tends to close the door structure **150**, the driving motor **166** drives the driver gear **1621** to rotate in the opposite direction, so that the door structure **150** can move in the opposite direction from the open position as shown in FIG. **6** to rotate and move to the closed position as shown in FIG. **4**. Under such configuration, when a temperature sensor of the portable electronic device detects that the temperature of the heating component **120** is higher than a warning temperature, the temperature sensor can send a heat-dissipation signal to the controller, and the controller may actuate the driving motor **166** accordingly, so as to automatically drive the door structure **150** to switch from the closed position to the open position to facilitate heat dissipation.

(24) In addition, the rotation and movement of the door structure **150** are driven by the driving

linkage **164** and the driven linkage **170** altogether, which can further increase the lifting height of the door structure **150**. With such configuration, referring to both FIG. **3** and FIG. **6**, when the door structure **150** is at the open position, a front side of the door structure **150** may form an air inlet opening OP**1** with the bracket **140**, and a rear side of the door structure **150** may form an air outlet opening OP**2** along with the bracket **140**, wherein the air inlet opening OP**1** faces the air outlet of the fan, so that the cooling airflow provided by the fan **220** may flow to the heat dissipation component **130** in the housing **110** through the air inlet opening OP**1**, and the hot airflow that performed heat exchange with the heat dissipation component **130** may flow out of the housing **110** through the air outlet opening OP**2** on the rear side, so as to improve the heat dissipation effect of the portable electronic device **100**.

(25) In other embodiments, the driving of the driving motor **166** can also cooperate with the fan module **200** to control opening and closing of the door structure **150**. For example, the portable electronic device **100** may include a sensing element (such as a magnetic sensor or a pressure/contact sensor, etc.), which is configured to sense whether the fan module **200** is mounted on the portable electronic device **100** or not. When the sensing element senses that the fan module **200** has been installed on the portable electronic device **100**, the sensing element may send a sensing signal to the controller, and the controller can activate the driving motor **166** accordingly to drive the door structure **150** to automatically switch from the closed position to the open position to facilitate heat dissipation. In this embodiment, the driving motor **166** can be locked onto the bracket **140** through fasteners, and then the bracket **140** is locked in the housing **110**, for example, the bottom plate in the housing **110**, along with the driving motor through the fasteners, so as to realize modular assembly.

(26) Referring to FIG. **7** to FIG. **10**, It must be noted here that the portable electronic device of the embodiment of FIG. **7** to FIG. **10** is similar to the portable electronic device of the preceding embodiment. The embodiment 10 uses the reference numerals and part of the content of the previous embodiments, wherein the same reference numerals are used to denote the same or similar components, and descriptions of the same technical content are omitted. For the description of the omitted part, reference may be made to the foregoing embodiments, and this embodiment will not be repeated. The difference between the portable electronic device of this embodiment and the portable electronic device of the foregoing embodiments will be described below.

(27) Referring to FIG. **7** to FIG. **10**, it is noted that the portable electronic device shown in FIG. **7** to FIG. **10** contains many features same as or similar to the portable electronic device disclosed earlier in the previous embodiments. For purpose of clarity and simplicity, detail description of same or similar features may be omitted, and the same or similar reference numbers denote the same or like components. The main differences between the portable electronic device in FIG. **7** to FIG. **10** and the portable electronic device in the previous embodiments are described as follows.

(28) In this embodiment, the opening and closing of the door structure **150***a* can be achieved manually. For example, referring to FIG. **7** and FIG. **8**, in some embodiments, the driving mechanism **160***a* may include a locking component **161**, which may include a handle **1611** and a locking portion **1612**. The handle **161** is configured to be pushed by an external force to drive the locking portion **1612** to move between a locking position as shown in FIG. **8** and a releasing position as shown in FIG. **10**. Correspondingly, the door structure **150***a* may include a hook **152** configured to be structurally interfered with the locking portion **1612**. In this way, when the locking portion **1612** is at the locking position, the locking portion **1612** is engaged with the hook **152** of the door structure **150***a*, so that the door structure **150***a* is fixed at the closed position as shown in FIG. **8**. In addition, the driving mechanism **160***a* may further include an elastic member **165**, which is coupled to and lean between the bracket **140** and the locking component **161**, so as to push the locking component **161** toward the hook **152** by its elastic restoring force, so as to maintain the locking relationship between the locking portion **1612** of the locking component **161** and the hook **152** of the door structure **150***a*.

(29) Please continue to refer to FIG. **8** to FIG. **10**, when the user wants to open the door structure **150***a*, an external force can be applied to the handle **1611** to make the locking component **161** move along the external force direction D**1**, e.g., the direction away from the hook **152**, so as to drive the locking portion **1612** to move away from the hook **152** to the release position as shown in FIG. **9** and FIG. **10**, so that the locking portion **1612** and the hook **152** are disengaged.
(30) In this embodiment, the driving mechanism **160***a* may further include an elastic restoring member **167**, which is coupled between the bracket **140** and the door structure **150***a*. When the door structure **150***a* is at the closed position as shown in FIG. **8**, the elastic restoring member **167** is compressed to generate an elastic restoring force. When the hook **152** is released, the elastic restoring member **167** utilizes its elastic restoring force to drive the door structure **150***a* to rotate and move upward. At this time, when the elastic restoring member **167** drives the door structure **150***a* to rotate and move, the door structure **150***a* also drives the driven linkage **170** to rotate and move, so that the elastic restoring member **167** and the driven linkage **170** jointly make the door structure **150***a* rotate and move to the open position as shown in FIG. **9**. Furthermore, the elastic restoring member **167** can be coupled to a first coupling point of the door structure **150***a*, and the driven linkage **170** can be coupled to a second coupling point of the door structure **150***a*, and the two coupling points can jointly control the rotating and moving path of the door structure **150***a*. In this way, when the door structure **150***a* is driven by the elastic restoring member **167** to rotate and being lifted, the door structure **150***a* will drive the driven linkage **170** coupled therewith to rotate and being lifted together, so that the elastic restoring member **167** and the driven linkage **170** can jointly control the rotating and moving path of the door structure **150** to make the door structure **150***a* to rotate and move between the closed position shown in FIG. **7** and FIG. **8** and the open position shown in FIG. **9** and FIG. **10**.
(31) Similarly, when the user wants to close the door structure **150***a*, the user may directly press the door structure **150***a* down till the hook **152** of the door structure **150***a* is engaged with the locking portion **1612** of the locking component **161**, such that the door structure **150***a* is returned from the open position shown in FIG. **9** and FIG. **10** to the closed position shown in FIG. **7** and FIG. **8**. In addition, the elastic member **165** leaning between the bracket **140** and the locking component **161** can utilize its elastic restoring force to push the locking component **161** toward the hook **152** to maintain the locking relationship between the locking portion **1612** of the locking component **161** and the hook **152** of the door structure **150***a*.
(32) To sum up, a portable electronic device of the disclosure includes a door structure and a housing having a heat dissipation opening, the heat dissipation opening is configured to expose at least a part of the heat dissipation component in the housing, and the door structure is driven by the driving mechanism and the driven linkage, which are pivotally connected between the bracket and the door structure, so as to cover or expose the heat dissipation opening. In addition, components such as the door structure, the driving mechanism and the driven linkage can be modularly assembled in the portable electronic device through the bracket. Moreover, the rotation and movement of the door structure is jointly driven by the driving linkage and the driven linkage of the driving mechanism, so as to further increase the lifting height of the door structure, thereby further improving the heat dissipation effect of the portable electronic device. With such configuration, when the portable electronic device has higher heat dissipation need, the door structure can be opened manually or automatically to facilitate heat dissipation. Therefore, the portable electronic device can have better heat dissipation efficiency, and the modular design can also simplify the complicated assembly steps. Moreover, the modular design enables the entire module to be tested, such as operation test, etc., before being assembled into the portable electronic device, so as to improve the product yield.
(33) It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications

and variations of this disclosure provided they fall within the scope of the following claims and their equivalents.

Claims

1. A portable electronic device, comprising: a housing comprising a heat dissipation opening; a heat dissipation component disposed in the housing and corresponding to the heat dissipation opening; a bracket disposed in the housing and surrounding the heat dissipation component; a door structure configured to rotate and move between a closed position covering the heat dissipation opening and an open position exposing the heat dissipation opening; and a driving mechanism pivotally connected between the bracket and the door structure to drive the door structure to rotate and move; and a driven linkage coupled between the bracket and the door structure, so as to drive the driven linkage to rotate and move when the door structure is driven to rotate and move, such that the driving mechanism and the driven linkage jointly drive the door structure to rotate and move between the closed position and the open position.
2. The portable electronic device as claimed in claim 1, wherein the driving mechanism comprises a gear set and a driving linkage coupled between the gear set and the door structure, and the gear set is configured to be driven to rotate, so as to drive the driving linkage to rotate and move, thereby drive the door structure to rotate and move.
3. The portable electronic device as claimed in claim 2, wherein the driving mechanism further comprises a driving motor, and the gear set further comprises a driver gear coupled to the driving motor and a idle gear coupled to the driver gear.
4. The portable electronic device as claimed in claim 3, wherein two opposite ends of the driving linkage are coupled to the idle gear and the door structure respectively.
5. The portable electronic device as claimed in claim 1, wherein the driving mechanism comprises a locking component comprising a handle and a locking portion, and the handle is configured to be pushed by an external force to drive the locking portion to move between a locking position and a releasing position.
6. The portable electronic device as claimed in claim 5, wherein the door structure comprises a hook, and when the locking portion is at the locking position, the locking portion is engaged with the hook, such that the door structure is fixed at the closed position, and when the handle is driven by the external force to move the locking position to the release position, the locking position and the hook are disengaged.
7. The portable electronic device as claimed in claim 6, wherein the driving mechanism further comprises an elastic restoring component coupled between the bracket and the door structure, when the door structure is at the closed position, the elastic restoring component is compressed to generate an elastic restoring force, and when the locking portion and the hook are disengaged, the elastic restoring component drive the door structure to rotate and move by the elastic restoring force.
8. The portable electronic device as claimed in claim 7, wherein when the elastic restoring component drives the door structure to rotate and move, the door structure drives the driven linkage to rotate and move, such that the elastic restoring component and the driven linkage jointly drive the door structure to rotate and move between the closed position and the open position.
9. The portable electronic device as claimed in claim 1, further comprising a fan module, including a frame and a fan disposed in the frame, wherein the frame is configured to be detachably disposed on the housing, and the frame comprises an opening corresponding to the heat dissipation opening.
10. The portable electronic device as claimed in claim 9, wherein an acute angle is included between a lower surface of the fan and a back surface of the housing, and an air outlet of the fan faces the heat dissipation opening.